
Index

- 1-of- K representation, 364
- ℓ_1 norm, 71
- ℓ_2 norm, 72
- abduction, 258
- Abel-Ruffini theorem, 334
- Abelian group, 36
- absolutely homogeneous, 71
- activation function, 315
- affine mapping, 63
- affine subspace, 61
- Akaike information criterion, 288
- algebra, 17
- algebraic multiplicity, 106
- analytic, 143
- ancestral sampling, 340, 364
- angle, 76
- associativity, 24, 26, 36
- attribute, 253
- augmented matrix, 29
- auto-encoder, 343
- automatic differentiation, 161
- automorphism, 49
- backpropagation, 159
- basic variable, 30
- basis, 44
- basis vector, 45
- Bayes factor, 287
- Bayes' law, 185
- Bayes' rule, 185
- Bayes' theorem, 185
- Bayesian GP-IVM, 347
- Bayesian inference, 274
- Bayesian information criterion, 288
- Bayesian linear regression, 303
- Bayesian model selection, 286
- Bayesian network, 278, 283
- Bayesian PCA, 346
- Bernoulli distribution, 205
- Beta distribution, 206
- bilinear mapping, 72
- bijective, 48
- binary classification, 370
- Binomial distribution, 206
- blind-source separation, 346
- Borel σ -algebra, 180
- canonical basis, 45
- canonical feature map, 389
- canonical link function, 315
- categorical variable, 180
- Cauchy-Schwarz inequality, 75
- change-of-variable technique, 219
- characteristic polynomial, 104
- Cholesky decomposition, 114
- Cholesky factor, 114
- Cholesky factorization, 114
- class, 370
- classification, 315
- closure, 36
- code, 343
- codirected, 105
- codomain, 58, 139
- collinear, 105
- column, 22
- column space, 59
- column vector, 22, 38
- completing the squares, 307
- concave function, 236
- condition number, 230
- conditional probability, 179
- conditionally independent, 195
- conjugate, 208
- conjugate prior, 208
- convex conjugate, 242
- convex function, 236
- convex hull, 386
- convex optimization problem, 236, 239
- convex set, 236
- coordinate, 50
- coordinate representation, 50
- coordinate vector, 50
- correlation, 191
- covariance, 190
- covariance matrix, 190, 198
- covariate, 253
- CP decomposition, 136
- cross-covariance, 191
- cross-validation, 258, 263
- cumulative distribution function, 178, 181
- d-separation, 281

- data covariance matrix, 318
- data point, 253
- data-fit term, 302
- decoder, 343
- deep auto-encoder, 347
- defective, 111
- denominator layout, 151
- derivative, 141
- design matrix, 294, 296
- determinant, 99
- diagonal matrix, 115
- diagonalizable, 116
- diagonalization, 116
- difference quotient, 141
- dimension, 45
- dimensionality reduction, 317
- directed graphical model, 278, 283
- direction, 61
- direction space, 61
- distance, 75
- distribution, 177
- distributivity, 24, 26
- domain, 58, 139
- dot product, 72
- dual SVM, 385
- Eckart-Young theorem, 131, 334
- eigendecomposition, 116
- eigenspace, 106
- eigspectrum, 106
- eigenvalue, 105
- eigenvalue equation, 105
- eigenvector, 105
- elementary transformations, 28
- EM algorithm, 360
- embarrassingly parallel, 264
- empirical covariance, 192
- empirical mean, 192
- empirical risk, 260
- empirical risk minimization, 257, 260
- encoder, 343
- endomorphism, 49
- epigraph, 236
- equivalent, 56
- error function, 294
- error term, 382
- Euclidean distance, 72, 75
- Euclidean norm, 72
- Euclidean vector space, 73
- event space, 175
- evidence, 186, 285, 306
- example, 253
- expected risk, 261
- expected value, 187
- exponential family, 205, 211
- extended Kalman filter, 170
- factor analysis, 346
- factor graph, 283
- feature, 253
- feature map, 254
- feature matrix, 296
- feature vector, 295
- Fisher discriminant analysis, 136
- Fisher-Neyman theorem, 210
- forward mode, 161
- free variable, 30
- full rank, 47
- full SVD, 128
- fundamental theorem of linear mappings, 60
- Gaussian elimination, 31
- Gaussian mixture model, 349
- Gaussian process, 316
- Gaussian process latent-variable model, 347
- general linear group, 37
- general solution, 28, 30
- generalized linear model, 272, 315
- generating set, 44
- generative process, 272, 286
- generator, 344
- geometric multiplicity, 108
- Givens rotation, 94
- global minimum, 225
- GP-LVM, 347
- gradient, 146
- Gram matrix, 389
- Gram-Schmidt orthogonalization, 89
- graphical model, 278
- group, 36
- Hadamard product, 23
- hard margin SVM, 377
- Hessian, 164
- Hessian eigenmaps, 136
- Hessian matrix, 165
- hinge loss, 381
- histogram, 369
- hyperparameter, 258
- hyperplane, 61, 62
- hyperprior, 281
- i.i.d., 195
- ICA, 346
- identity automorphism, 49
- identity mapping, 49
- identity matrix, 23
- image, 58, 139
- independent and identically distributed, 195, 260, 266
- independent component analysis, 346
- inference network, 344
- injective, 48
- inner product, 73
- inner product space, 73
- intermediate variables, 162
- inverse, 24

- inverse element, 36
- invertible, 24
- Isomap, 136
- isomorphism, 49
- Jacobian, 146, 150
- Jacobian determinant, 152
- Jeffreys-Lindley paradox, 287
- Jensen's inequality, 239
- joint probability, 178
- Karhunen-Loève transform, 318
- kernel, 33, 47, 58, 254, 388
- kernel density estimation, 369
- kernel matrix, 389
- kernel PCA, 347
- kernel trick, 316, 347, 389
- label, 253
- Lagrange multiplier, 234
- Lagrangian, 234
- Lagrangian dual problem, 234
- Laplace approximation, 170
- Laplace expansion, 102
- Laplacian eigenmaps, 136
- LASSO, 303, 316
- latent variable, 275
- law, 177, 181
- law of total variance, 203
- leading coefficient, 30
- least-squares loss, 154
- least-squares problem, 261
- least-squares solution, 88
- left-singular vectors, 119
- Legendre transform, 242
- Legendre-Fenchel transform, 242
- length, 71
- likelihood, 185, 265, 269, 291
- line, 61, 82
- linear combination, 40
- linear manifold, 61
- linear mapping, 48
- linear program, 239
- linear subspace, 39
- linear transformation, 48
- linearly dependent, 40
- linearly independent, 40
- link function, 272
- loading, 322
- local minimum, 225
- log-partition function, 211
- logistic regression, 315
- logistic sigmoid, 315
- loss function, 260, 381
- loss term, 382
- lower-triangular matrix, 101
- Maclaurin series, 143
- Manhattan norm, 71
- MAP, 300
- MAP estimation, 269
- margin, 374
- marginal, 190
- marginal likelihood, 186, 286, 306
- marginal probability, 179
- marginalization property, 184
- Markov random field, 283
- matrix, 22
- matrix factorization, 98
- maximum a posteriori, 300
- maximum a posteriori estimation, 269
- maximum likelihood, 257
- maximum likelihood estimate, 296
- maximum likelihood estimation, 265, 293
- mean, 187
- mean function, 309
- mean vector, 198
- measure, 180
- median, 188
- metric, 76
- minimal, 44
- minimax inequality, 234
- misfit term, 302
- mixture model, 349
- mixture weight, 349
- mode, 188
- model, 251
- model evidence, 286
- model selection, 258
- Moore-Penrose pseudo-inverse, 35
- multidimensional scaling, 136
- multiplication by scalars, 37
- multivariate, 178
- multivariate Gaussian distribution, 198
- multivariate Taylor series, 166
- natural parameters, 212
- negative log-likelihood, 265
- nested cross-validation, 258, 284
- neutral element, 36
- noninvertible, 24
- nonsingular, 24
- norm, 71
- normal distribution, 197
- normal equation, 86
- normal vector, 80
- null space, 33, 47, 58
- numerator layout, 150
- Occam's razor, 285
- ONB, 79
- one-hot encoding, 364
- ordered basis, 50
- orthogonal, 77
- orthogonal basis, 79
- orthogonal complement, 79
- orthogonal matrix, 78
- orthonormal, 77
- orthonormal basis, 79

- outer product, 38
- overfitting, 262, 271, 299
- PageRank, 114
- parameters, 61
- parametric equation, 61
- partial derivative, 146
- particular solution, 27, 30
- PCA, 317
- pdf, 181
- penalty term, 263
- pivot, 30
- plane, 62
- plate, 281
- population mean and covariance, 191
- positive definite, 71, 73, 74, 76
- posterior, 185, 269
- posterior odds, 287
- power iteration, 334
- power series representation, 145
- PPCA, 340
- preconditioner, 230
- predictor, 12, 255
- primal problem, 234
- principal component, 322
- principal component analysis, 136, 317
- principal subspace, 327
- prior, 185, 269
- prior odds, 287
- probabilistic inverse, 186
- probabilistic PCA, 340
- probabilistic programming, 278
- probability, 175
- probability density function, 181
- probability distribution, 172
- probability integral transform, 217
- probability mass function, 178
- product rule, 184
- projection, 82
- projection error, 88
- projection matrix, 82
- pseudo-inverse, 86
- random variable, 172, 175
- range, 58
- rank, 47
- rank deficient, 47
- rank- k approximation, 130
- rank-nullity theorem, 60
- raw-score formula for variance, 193
- recognition network, 344
- reconstruction error, 88, 327
- reduced hull, 388
- reduced row-echelon form, 31
- reduced SVD, 129
- REF, 30
- regression, 289
- regular, 24
- regularization, 262, 302, 382
- regularization parameter, 263, 302, 380
- regularized least squares, 302
- regularizer, 263, 302, 380, 382
- representer theorem, 384
- responsibility, 352
- reverse mode, 161
- right-singular vectors, 119
- RMSE, 298
- root mean square error, 298
- rotation, 91
- rotation matrix, 92
- row, 22
- row vector, 22, 38
- row-echelon form, 30
- sample mean, 192
- sample space, 175
- scalar, 37
- scalar product, 72
- sigmoid, 213
- similar, 56
- singular, 24
- singular value decomposition, 119
- singular value equation, 124
- singular value matrix, 119
- singular values, 119
- slack variable, 379
- soft margin SVM, 379, 380
- solution, 20
- span, 44
- special solution, 27
- spectral clustering, 136
- spectral norm, 131
- spectral theorem, 111
- spectrum, 106
- square matrix, 25
- standard basis, 45
- standard deviation, 190
- standard normal distribution, 198
- standardization, 336
- statistical independence, 194
- statistical learning theory, 265
- stochastic gradient descent, 231
- strong duality, 236
- sufficient statistics, 210
- sum rule, 184
- support point, 61
- support vector, 384
- supporting hyperplane, 242
- surjective, 48
- SVD, 119
- SVD theorem, 119
- symmetric, 73, 76
- symmetric matrix, 25
- symmetric, positive definite, 74
- symmetric, positive semidefinite, 74
- system of linear equations, 20
- target space, 175

Taylor polynomial, 142, 166
Taylor series, 142
test error, 300
test set, 262, 284
Tikhonov regularization, 265
trace, 103
training, 12
training error, 300
training set, 260, 292
transfer function, 315
transformation matrix, 51
translation vector, 63
transpose, 25, 38
triangle inequality, 71, 76
truncated SVD, 129
Tucker decomposition, 136
underfitting, 271
undirected graphical model, 283
uniform distribution, 182
univariate, 178
unscented transform, 170
upper-triangular matrix, 101
validation set, 263, 284
variable selection, 316
variance, 190
vector, 37
vector addition, 37
vector space, 37
vector space homomorphism, 48
vector space with inner product, 73
vector subspace, 39
weak duality, 235
zero-one loss, 381